PRE-ALGEBRA (8TH) · GEOMETRY

Pythagorean Squares Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Pythagorean Squares Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A right triangle has legs 3 and 4. What are the areas of the squares on the legs? Predict the area of the square on the hypotenuse.

2. Does $a^2 + b^2 = c^2$ work for triangles that are not right triangles?

2 Explore use the tool

1. Make a right triangle with legs 6 and 8. Record the three square areas.

2. Make a triangle with an angle greater than 90°. Compare $a^2 + b^2$ with c^2.

3. Make a right triangle with no horizontal or vertical side. Use the tilted-square trick on the biggest square and show the calculation.

4. Make $a^2 + b^2$ exactly 2 more than c^2. Is the angle opposite c exactly 90°?

3 Record use the tool and fill in the table

Square areas

Every row can be built on the grid. a^2 and b^2 are the two smaller squares, c^2 the biggest.

a^2	b^2	c^2	$a^2 + b^2$? c^2	Type of triangle
9	16	25		
5	16	29		
2	5	5		
8	18	26		
1	1			

4 Explain what did you discover?

1. In a right triangle, $a^2 + b^2$ _____ c^2, where c is the _____.
2. If $a^2 + b^2 > c^2$, the triangle is _____. If $a^2 + b^2 < c^2$, it is _____.
3. To find c, take the _____ of $a^2 + b^2$.
4. Kai says a triangle with sides 5, 6 and 8 is a right triangle because it looks close to a 6-8-10 triangle. Use squares to check.

5 Apply a new situation, no tool

1. A 13-ft ladder reaches 12 ft up a wall. How far is the base of the ladder from the wall?
2. Find the distance between (1, 2) and (7, 10).
3. Is a triangle with sides 9, 40 and 41 a right triangle? Show why.
4. A square has a diagonal of 10 cm. Find its side length, exactly and to the nearest tenth.